

Process Integration of a 27nm, 16Gb Cu ReRAM

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Introduction:

Resistive RAM (ReRAM) is a promising candidate for future non volatile memory applications. Despite an abundance of literature on resistive memory cells with numerous materials (1), demonstration of high density memories suitable for relevant commercial applications is absent. We have proposed the viability of a Cu-based ReRAM Cell capable of meeting storage class memory specifications (2, 3). This paper describes a 16Gb Non-volatile (NV) Cu-based ReRAM chip fabricated at the 27nm node. 180MB/s write and 900MB/s read performance is enabled. The array architecture is 6F2 and utilizes a highly modified BRAD (Buried Recessed Access Device) coupled with novel, self-aligned plugs contacting a Damascene Cu-ReRAM Cell. Line-SAC (Self Aligned Contact) Digit lines running over the array complete cell formation.

Cell Architecture:

ReRAM cell technology requires an access device to enable sufficiently large array blocks (4). Therefore we have leveraged 27nm 6F2 DRAM array technology as a starting point, which yields a $0.0044\mu\text{m}^2$ cell. Pitch multiplication is required for multiple levels to reach 27nm geometries. Bipolar ReRAM operation swings the cell plate bias for Program/Erase operations which requires patterning the cell plate into stripes called Common Source Plate (CSP), Fig 1.

Establishing planarity between the array and periphery circuitry is accomplished by building the arrays in blocks of raised epitaxial silicon, Fig 2. Silicon grown within confined regions typically suffers from high defect levels. We control defects within the epitaxial regions using a carefully engineered stress relief boundary accompanied by

a small perimeter circuit “keep out” zone (or hedge). The thickness of Si epitaxy is chosen to align for planarization with the periphery CMOS gate stack.

DRAM access devices are engineered for extremely low leakage levels. However, the ReRAM cell places a much different set of requirements on access device performance; namely significantly higher drive with moderate I_{off} .

Access Device (A-FET):

While the A-FET for ReRAM superficially resembles a DRAM BRAD, significant modifications are required to deliver sufficient programming current. First, by relaxing the DRAM ultra-low I_{off} requirement, silicide can be introduced into the array to minimize pillar contact resistance. Silicide formation is challenging and must be carefully engineered to form at the bottom of the Exhumed Contact opening with limited reactive Si availability.

Modifications were also made to the A-FET Si trench and saddle profiles. Increasing the saddle height of the device while limiting the silicon trench depth serves to both minimize parasitic resistances and increase the effective fin ratio of the buried device. Optimizing the profile and ratio of these dimensions is a strong driver of device performance.

The gate oxide was developed to support elevated array voltages and suppress leakage current. Finally, implants and thermal budgets were re-designed to align with the new physical device structure and to maximize activation in the confined pillar. Completed A-FET images are shown in Fig 3.

Typical A-FET electrical performance is shown in Fig 4a-b. Figure 4c illustrates performance improvements derived from the multiple optimizations discussed above which culminates in I_{on} of 40 μ A/device. As shown in the top curves of Fig 4c, test structure results are very well aligned with actual array curves. Finally, Figure 4d demonstrates current distributions taken from the array through 1M cycles under various bias conditions.

Array Contacts:

Contacting the S/D of the A-FET in the array using conventional techniques requires an orthogonal, double pitch multiplication process. Such an approach is burdened with poor registration margin and high process costs, so we have introduced a novel self-aligned contact. This contact is formed by the exhumation of a sacrificial layer which is introduced into the stack prior to STI formation. During STI and A-FET structure formation, the sacrificial layer is chopped to form an isolated plug which is then selectively removed. As described previously, contact silicide is critical for enhancing AFET drive capability. Our flow utilizes a single silicide process to form both array and periphery contacts simultaneously. Materials for the bottom electrode (BE) contacting the cell are selected to enhance cell performance. The BE is deposited on the silicide and then isolated using CMP prior to Cell Deposition, Fig 5.

Cell Formation:

ReRAM cells are sensitive to contamination and damage which can be introduced during a subtractive cell process; i.e. reactive ion etching. These process related defects manifest themselves as time zero fails, increased cell variability, and degraded reliability. Introduction of a Cu-Damascene cell module circumvents these issues by removing the switching regions of the cell from proximity to process related defects. The Damascene trench is filled with an electrolyte layer, a Cu-based ion reservoir, and a metal strapping layer. Nitride encapsulation of the cell metal strap and sidewalls is critical for both cell functionality and isolation of the cell from the subsequent bit line formation.

Set and reset distributions from the 16Gb array are shown in Fig 6 and BER in Table 2. We note that 16Gb array cell distributions are shown to be comparable to that of both an intrinsic cell (planar “edgeless” cell) as well as a “scaled cell” (damascene cell at relaxed pitch). Equivalent performance

from the three different cell structures demonstrates a successful integration flow whereby the intrinsic cell performance is translated directly into dense array structures with no loss of capability.

Bit Line (BL) Formation:

Bit lines are formed using an industry-first Line-SAC etch process over the Damascene Cell, which eliminates the need for a separate plug contact layer (Fig 7). Use of an etch stop layer in the periphery allows the array BL metal to also serve as a local interconnect in the periphery. Metal fill is very challenging at the array geometries ($AR > 10:1$) and is enabled with an aggressive Cu Damascene process. Resistance meets design targets of 800m Ω /sq.

Chip Performance:

The chip design consists of an 8-bank interleaved architecture with DDR interface (2). Currently we demonstrate a throughput of 900MB/s for read and 180MB/s for write operations. These results are shown in Table 1. The on-chip programmable controller algorithm allows great flexibility in balancing cell functionality and reliability with chip performance. Ongoing cell and algorithm improvements are expected to enable throughput at peak design capability.

Summary:

A 27nm 16Gb Cu based NV Re-RAM chip has been demonstrated. Novel process introduction to enable this technology include a Damascene Cell, Line-SAC Digit Lines filled with Cu, exhumed-silicided array contacts, raised epitaxial arrays, and high-drive buried access devices.

References:

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- (2) R. Fackenthal, Makoto Kitagawa, et al, “*A 16Gb ReRAM with 200MB/s Write and 1GB/s Read in 27nm Technology*”, ISSCC 2014, pp. 21-22
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- (4) G. Burr, R. Shenoy, K. Virwani, P. Narayanan, A Padilla, B. Kurdi, H. Hwang, “*Access devices for 3D crosspoint memory*”, J. Vac Sci. Technol. B 32(4), Jul/Aug 2014

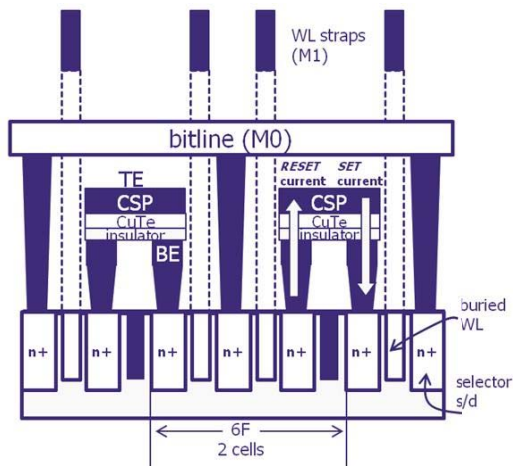


Figure 1: Schematic of Cu-ReRAM 6F2 Architecture (1)

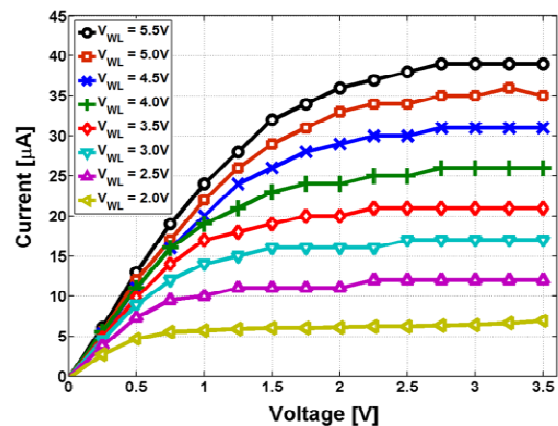


Figure 2: 3D Perspective showing Epi Array used for planarity

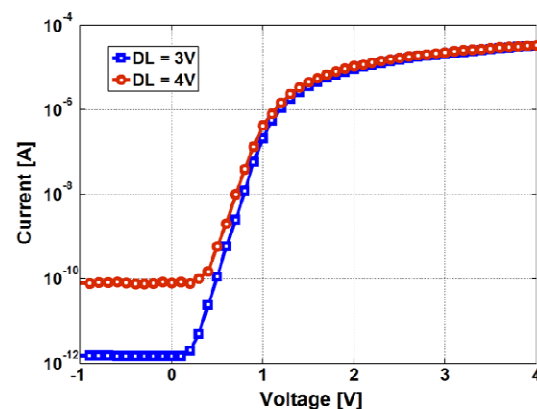


Figure 3: Completed 27nm array (x-direction) showing A-FET, silicide, metal plugs, Damascene cell, and SAC DL.

Figure 4a-b: A-FET IV curves. SS=150mV/dec

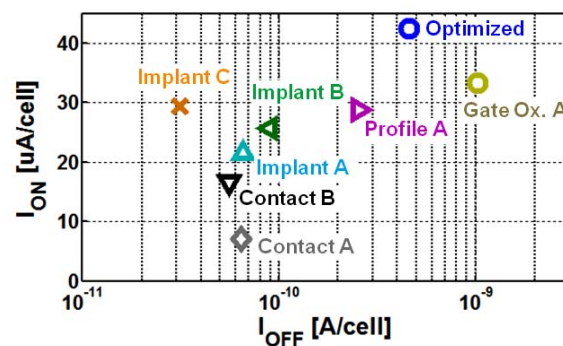
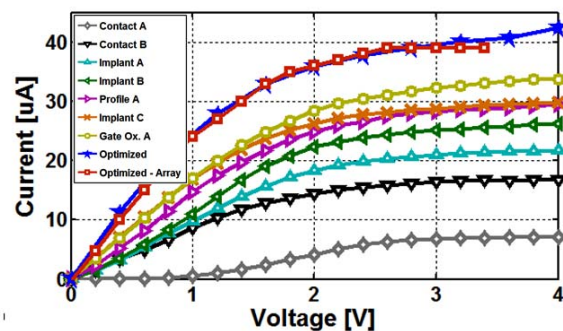


Figure 5: Exhumed Contact showing silicide and Metal Plug completed prior to Cell deposition.

Figure 4c: A-FET performance improvements through contact, structure, implant, and gate oxide optimization. Note alignment of test structure and array results seen in top transistor curve overlay.

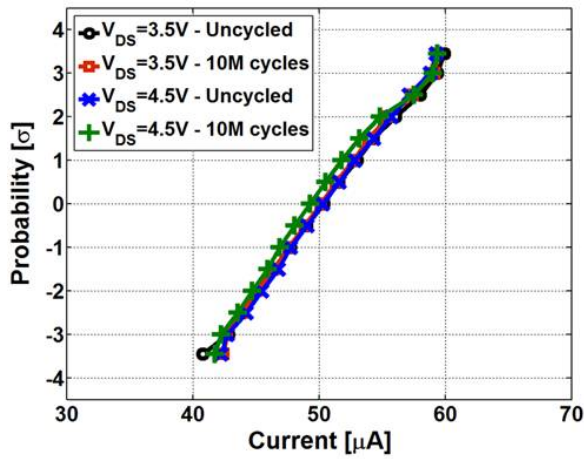


Figure 4d: A-FET array current distributions through 1M cycles.

($V_{GS} = 5.5V$)

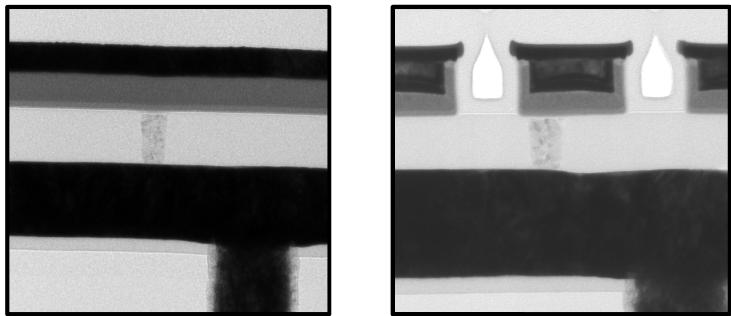


Figure 6: Cell Structures used in comparisons of cell performance: Intrinsic Edgeless Cell (left) vs. Scaled isolated Cell (right) vs. Full Integrated Cell shown in Figure 3.

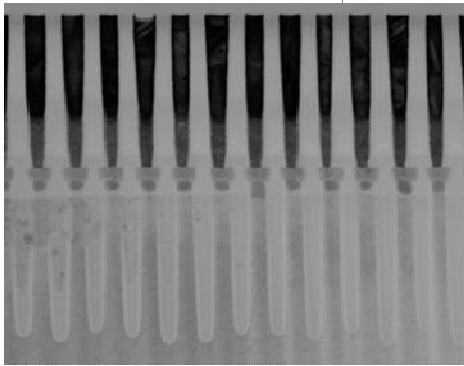


Figure 7: Cross-section of array in the y-direction showing DL runners and high aspect ratio Cu fill.

Table 1: Chip performance comparison of design target with silicon results to date.

Summary Table			
Feature		Design Capability	Actual Demonstration
Density		16 Gb	16 Gb
Tech node (nm)		27	27
Cell Size (nm ²)		4374 (6F ²)	4374 (6F2)
Die Size (mm ²)		168	168
Selector		Buried WL MOS selector	Buried WL MOS selector
Read Performance	BW (MB/s)	1000	900
	Latency (μs)	2	2.3
Write Performance	BW (MB/s)	200	180
	Latency (μs)	10	11.7

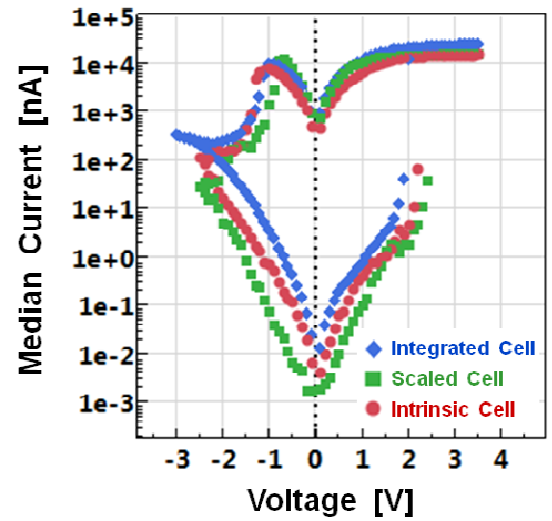


Figure 6a: DC IV curves comparing Intrinsic Cell (Planar edgeless cell), Scaled Cell (Damascene Isolated Cell), and the fully Integrated Cell at 27nm

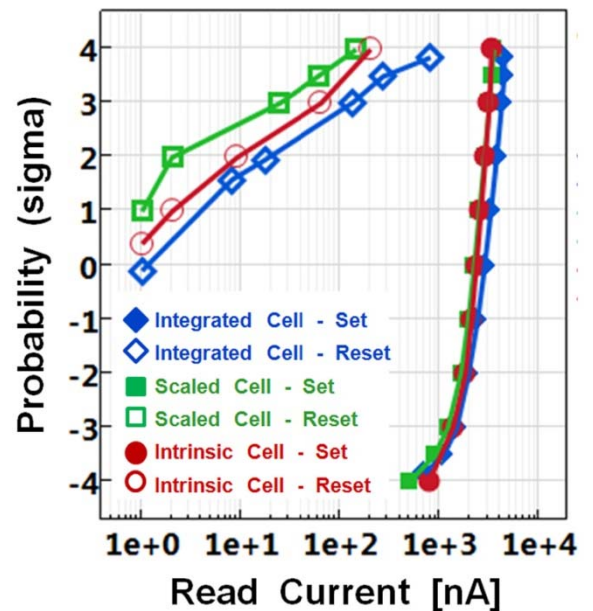


Figure 6b: Cell Current distributions after 1k cycles showing comparable behavior.

Table 2: Cell BER summary through cycling and retention bake.

Cycle #	BER Cycling	BER Cycle + Bake*
10^3	8.9E-05	1.4E-03
10^4	1.2E-04	8.9E-04
10^5	8.9E-05	2.2E-03

* 1yr 55°C Retention BER ($E_A = 1.5eV$)